Institute of Engineering & Technology, Indore
BE-II (Electronics & Instrumentation Engg.)
Class Test-I (Aug.' 2023)
3AERC1: APPLIED MATHEMATICS III

Time: 70 min **Max Marks: 20**

Note: Attempt any FOUR questions. Each question carries equal marks.

Q.1 Find the real root of the equation $2x - log_{10}x = 7$ by Regula Falsi method correct to five decimal places.

Q.2 Solve the system of equation by using Jacobi's Iteration method upto the end of sixth iteration: $10x-2y-3z=205$, $2x-10y+2z=-154$, $2x+y-10z=-120$

Q.3 Find the first and tenth term of the series:

x:	3	4	5	6	7	8	9
F(x):	2.7	6.4	12.5	21.6	34.3	51.2	72.9

Q.4 Evaluate $\Delta^2\left[\frac{5x+12}{x^2+5x+6}\right]$ interval of differencing being unity.

Q.5 The following table gives population of a town during the last six censuses.

Year	1941	1951	1961	1971	1981	1991
Population in 10 thousand	12	15	20	27	39	52

Estimate the increase in population from 1976 to 1978.

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(e) Solve by method ... given that $y(0) = 0,\ y'(0) = 0,\ y''(0) = 6.$

IET, DAVV, Indore
B.E. II (Civil)
Class Test I (August-2023)
3AVRC1: Applied Mathematics-III

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Q.1. Evaluate (05)

a) $L\{\, t \int_0^t e^{-t} \frac{\sin t}{t}\, dt \,\}$ b) $\int_0^\infty e^{-t} \left(\frac{\cos at - \cos bt}{t}\right) dt$

Q.2. Using convolution theorem to find: (05)

a) $L^{-1}\{ \frac{1}{s^2(s+1)^2} \}$ b) $L^{-1}\{ \frac{s^2}{(s+4)^2} \}$

Q.3. Using the method Laplace transform, solve the equation: $(D^2 - 3D + 2)\, y = 1 - e^{2t}$ (05)

Q.4. Obtain a root of equation **$\cos x = xe^x$** using **Bisection method,** correct to three decimal places. (05)

Q.5. Find a real root of the equation **$x \log_{10} x = 1.2$** by **Regula falsi method,** correct to four decimal places. (05)

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